

Name: Saidjon Nozimboev

Student ID: 110186060

1)

1.e $(\sim r \wedge q) \vee (p \wedge q) \vee (\sim p \wedge \sim r)$

p	q	r	$\sim r \wedge q$	$p \wedge q$	$\sim p \wedge \sim r$	Final
T	T	T	F	T	F	T
T	T	F	T	T	F	T
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	F	F	F
F	T	F	T	F	T	T
F	F	T	F	F	F	F
F	F	F	F	F	T	T

2)

3 (b)

$$B \vee \sim (a \Rightarrow (B \vee \gamma)) \vee (a \Rightarrow \gamma)$$

1 $\sim (B \vee \sim (a \Rightarrow (B \vee \gamma)) \vee (a \Rightarrow \gamma))$ hypothesis

2 $\sim B \wedge \sim \sim (a \Rightarrow (B \vee \gamma)) \wedge \sim (a \Rightarrow \gamma)$ 1, EI7

3	$\sim B \wedge (a \Rightarrow (B \vee \gamma)) \wedge \sim(a \Rightarrow \gamma)$	2, E15
4	$\sim B$	3, I2
5	$a \Rightarrow (B \vee \gamma)$	3, E12
6	$\sim(a \Rightarrow \gamma)$	3, I2
7	$\sim(\sim a \vee \gamma)$	6, E18
8	$a \wedge \sim \gamma$	7, E17, E15
9	a	8, I2
10	$\sim \gamma$	8, I2
11	$\sim B \wedge \sim \gamma$	4, 10, I6
12	$\sim(B \vee \gamma)$	11, E17
13	$\sim a$	5, 12, I4
14	$a \wedge \sim a$	9, 13, I6
15	False	14, E1

Hence, $\vdash \beta \vee \sim(a \Rightarrow (B \vee \gamma)) \vee (a \Rightarrow \gamma)$

(h)

$$(a \Rightarrow (\beta \wedge \sim \beta)) \Rightarrow \sim a$$

$$\perp \sim((a \Rightarrow (\beta \wedge \sim \beta)) \Rightarrow \sim a) \text{ hypothesis}$$

2	$\sim(\sim(a \Rightarrow (B \wedge \sim B)) \vee \sim a)$	1, E18
3	$\sim\sim(a \Rightarrow (B \wedge \sim B)) \wedge \sim\sim a$	2, E17
4	$(a \Rightarrow (B \wedge \sim B)) \wedge a$	3, E15
5	$a \Rightarrow (B \wedge \sim B)$	4, I2
6	$a \wedge (a \Rightarrow (B \wedge \sim B))$	4, E9
7	a	6, I2
8	$B \wedge \sim B$	5, 7 I3
9	False	8, E1

Hence, $\vdash(a \Rightarrow (B \wedge \sim B)) \Rightarrow \sim a$

(k)

$$a \Rightarrow (B \Leftrightarrow B)$$

1	$\sim(a \Rightarrow (B \Leftrightarrow B))$	hypothesis
2	$\sim(\sim(a \vee (B \Leftrightarrow B)))$	1, E18
3	$\sim\sim a \wedge \sim(B \Leftrightarrow B)$	2, E17
4	$a \wedge \sim(B \Leftrightarrow B)$	3, E15
5	a	4, I2
6	$\sim(B \Leftrightarrow B)$	4, E9, I2
7	$\sim((B \Rightarrow B) \wedge (B \Rightarrow B))$	6, E20

8	$\sim (B \Rightarrow B) \vee \sim (B \Rightarrow B)$	7, E16
9	$\sim (\sim B \vee B) \vee \sim (\sim B \vee B)$	8, E18
10	$(\sim \sim B \wedge \sim B) \vee (\sim \sim B \wedge \sim B)$	9, E17
11	$(B \wedge \sim B) \vee (B \wedge \sim B)$	10, E15
12	false $\vee$ false	11, E1
13	false	12, E6

Hence, $\vdash a \Rightarrow (B \Leftrightarrow B)$

3) 6 (b)

$$\begin{array}{l}
 P_1 : (p \vee q) \Rightarrow (r \wedge s) \\
 P_2 : (s \vee t) \Rightarrow u \\
 \hline
 C : p \Rightarrow u
 \end{array}$$

- 1 $(p \vee q) \Rightarrow (r \wedge s)$ From Γ
- 2 $(s \vee t) \Rightarrow u$ from Γ
- 3 $\sim (p \vee q) \vee (r \wedge s)$ 1, E18
- 4 $(\sim p \wedge \sim q) \vee (r \wedge s)$ 3, E17
- 5 $((\sim p \wedge \sim q) \vee s) \wedge ((\sim p \wedge \sim q) \wedge (\sim q \vee r))$ 4, E14, 10
- 6 $((\sim p \vee s) \wedge (\sim q \vee s)) \wedge (\sim p \vee r) \wedge (\sim q \vee r)$ 5, E14, 10

7	$(\sim p \vee s) \wedge (\sim q \vee s)$	6, I2
8	$\sim p \vee s$	7, I2
9	$p \Rightarrow s$	8, E18
10	$\sim (s \vee t) \vee u$	2, E18
11	$(\sim s \wedge \sim t) \vee u$	10, E17
12	$u \vee (\sim s \wedge \sim t)$	11, E10
13	$(u \vee \sim s) \wedge (u \vee \sim t)$	12, E14
14	$(\sim s \vee u) \wedge (\sim t \vee u)$	13, E10
15	$\neg s \vee u$	14, I2
16	$s \Rightarrow u$	15, E18
17	$p \Rightarrow u$	9, 16, I5

Hence, $P_1, P_2 \vdash p \Rightarrow u$

(9)

$$P1: p \wedge q \Rightarrow r$$

$$P2: p \vee q \Rightarrow u$$

$$P3: r \Rightarrow \sim u$$

$$C: p \Rightarrow \sim q$$

- 1 $p \wedge q \Rightarrow r$ From Γ
- 2 $p \vee q \Rightarrow u$ From Γ
- 3 $r \Rightarrow \sim u$ From Γ
- 4 $(p \wedge q) \Rightarrow \sim u$ 1,3 I5
- 5 $\sim \sim u \Rightarrow \sim(p \wedge q)$ 4, E19
- 6 $u \Rightarrow \sim(p \wedge q)$ 5, E15
- 7 $(p \vee q) \Rightarrow \sim(p \wedge q)$ 2,6, I5
- 8 $\sim(p \vee q) \vee \sim(p \wedge q)$ 7, E18
- 9 $(\sim p \wedge \sim q) \vee (\sim p \vee \sim q)$ 8, E16, I7
- 10 $((\sim p \wedge \sim q) \vee \sim p) \vee \sim q$ 9, E12
- 11 $(\sim p \vee (\sim p \vee \sim q)) \vee \sim q$ 10, E10
- 12 $((\sim p \wedge \text{true}) \vee (\sim p \wedge \sim q)) \vee \sim q$ 11, E5
- 13 $(\sim p \wedge (\text{true} \vee \sim q)) \vee \sim q$ 12, E13
- 14 $(\sim p \wedge \text{true}) \vee \sim q$ 13, E8
- 15 $\sim p \vee \sim p$ 14, E5
- 16 $p \Rightarrow \sim q$ 15, E18

Hence, $P1, P2, P3 \vdash p \Rightarrow \sim q$

(11) $P1: (p \Rightarrow q) \wedge (r \Rightarrow s)$

$P2: (q \Rightarrow t) \wedge (s \Rightarrow u)$

$P3: \sim (t \wedge u)$

$P4: p \Rightarrow r$

$C: \sim p$

1 $(p \Rightarrow q) \wedge (r \Rightarrow s)$ From Γ

2 $(q \Rightarrow t) \wedge (s \Rightarrow u)$ From Γ

3 $\sim (t \wedge u)$ From Γ

4 $p \Rightarrow r$ From Γ

5 $p \Rightarrow q$ 1, I2

6 $r \Rightarrow s$ 1, E9, I2

7 $q \Rightarrow t$ 2, I2

8 $s \Rightarrow u$ 2, E9, I2

9 $p \Rightarrow t$ 5, 7, I5

10 $p \Rightarrow s$ 4, 6, I5

11 $p \Rightarrow u$ 10, 8, I5

12 $\sim p \vee t$ 9, E18

13 $\sim p \vee u$ 11, E18

14	$(\sim p \vee t) \wedge (\sim p \vee u)$	12, 13, I6
15	$\sim p \vee (t \wedge u)$	14, E14
16	$p \Rightarrow (t \wedge u)$	15, E18
17	$\sim p$	3, 16, I4

Hence, $p_1, p_2, p_3, p_4 \vdash \sim p$

4)

7 (c)

Let

c: the contrast is satisfied

b: the building is completed by Nov 30

e: the electrical subcontractor completes the work by November 10

l: the bank loses money

$$p_1: c \Leftrightarrow b$$

$$p_2: e \Leftrightarrow b$$

$$p_3: l \Leftrightarrow \sim c$$

$$C: e \Leftrightarrow \sim l$$

$$1 \quad c \Leftrightarrow b$$

From Γ

$$2 \quad e \Leftrightarrow b$$

From Γ

$$3 \quad l \Leftrightarrow \sim c$$

From Γ

4	$(c \Rightarrow b) \wedge (b \Rightarrow c)$	1, E20
5	$c \Rightarrow b$	4, I2
6	$b \Rightarrow c$	4, E9, I2
7	$(e \Rightarrow b) \wedge (b \Rightarrow e)$	2, E20
8	$e \Rightarrow b$	7, I2
9	$b \Rightarrow e$	7, E9, I2
10	$(l \Rightarrow \sim c) \wedge (\sim c \Rightarrow l)$	3, E20
11	$l \Rightarrow \sim c$	10, I2
12	$\sim c \Rightarrow l$	10, E9, I2
13	$\sim \sim c \Rightarrow \sim l$	11, E19
14	$c \Rightarrow \sim l$	13, E15
15	$\sim l \Rightarrow \sim \sim c$	12, E19
16	$\sim l \Rightarrow c$	15, E15
17	$e \Rightarrow c$	8, 6, I5
18	$e \Rightarrow \sim l$	17, 14, I5
19	$\sim l \Rightarrow b$	16, 5, I5
20	$\sim l \Rightarrow e$	19, 9, I5
21	$(e \Rightarrow \sim l) \wedge (\sim l \Rightarrow e)$	18, 20, I6
22	$e \Leftrightarrow l$	21, E20

Hence,

$p_1, p_2, p_3 \vdash e \Leftrightarrow \sim l$